1. Bisect an 80 mm long line.
2. Bisect a circular arc
3. Draw an angle of 75^0 and bisect it with the help of a compass.
4. Trisect a right angle.
5. Divide an 80 mm long straight line into seven equal parts
6. Divide a 50 mm diameter circle into 12 equal segments.
7. Construct a Triangle of side 30 mm when a side is
   (a) Parallel to the ground
   (b) Perpendicular to the ground
   (c) Inclined at an angle of 30^0 to the ground
   (d) All sides are equally inclined to the ground
8. Construct a Square of side 30 mm when a side is
   (a) Parallel to the ground
   (b) Perpendicular to the ground
   (c) Inclined at an angle of 30^0 to the ground
   (d) All sides are equally inclined to the ground
9. Construct a Pentagon of side 30 mm when a side is
   (a) Parallel to the ground
   (b) Perpendicular to the ground
   (c) Inclined at an angle of 30^0 to the ground
   (d) All sides are equally inclined to the ground
10. Construct a Hexagon of side 30 mm when a side is
   (a) Parallel to the ground
   (b) Perpendicular to the ground
   (c) Inclined at an angle of 30^0 to the ground
   (d) All sides are equally inclined to the ground

11. Inscribe a Circle in a Triangle of 60 mm side.
12. Inscribe a Circle in a Square of 50 mm side.
13. Inscribe a Circle in a Pentagon of 40 mm side.
14. Inscribe a Circle in a Hexagon of 30 mm side.
15. Inscribe a Circle of 40 mm diameter in a Triangle.
16. Inscribe a Circle of 40 mm diameter in a Square.
17. Inscribe a Circle of 40 mm diameter in a Pentagon.
18. Inscribe a Circle of 40 mm diameter in a Hexagon.
19. Circumscribe a Circle of 60 mm diameter to a Triangle.
20. Circumscribe a Circle of 60 mm diameter to a Square.
21. Circumscribe a Circle of 60 mm diameter to a Pentagon.
22. Circumscribe a Circle of 60 mm diameter to a Hexagon.
23. Circumscribe a Circle to a Triangle of 60 mm side.
24. Circumscribe a Circle to a Square of 50 mm side.
25. Circumscribe a Circle to a Pentagon of 40 mm side.
26. Circumscribe a Circle to a Hexagon of 30 mm side.

UNIT – I

PLAIN SCALES

1. Construct a scale of 1 : 40 to read meters and decimeters and long enough to measure up to 6 meters. Mark a distance of 4.7 m on it.
2. If 1 centimeter long line on a map represents a real length of 4 meters. Calculate the R.F and draw a plain scale long enough to measure up to 50 meters. Show a distance of 44 m on it.
3. A real length of 1 decameter is represented by a line of 5 cm in a drawing. Find the R.F and construct a plain scale to measure up to 2.5 decameters. Mark a distance of 19 m on it.
4. Construct a scale of 1:5 to show decimeters and centimeters and long enough to measure up to 1 m. Show a distance of 6.3 dm on it.
5. A line of 1 centimeter represents an actual length of 4 dm. Draw a plain scale and mark a distance of 6.7 m on it.
6. An area of 49 square centimeters on a map represents an area of 16 sq. m on a field. Draw a scale long enough to measure 8 m. Mark a distance of 6 m 9 dm on the scale.
7. A cube of 5 cm side represents a tank of 1000 cubic meters volume. Find the R.F and construct a scale to measure up to 35 m. Mark a distance of 27 m on it.
8. Construct a scale of 1 : 14 to read feet and inches and long enough to measure 7 feet. Show a distance of 5 feet 10 inches on it.
9. Construct a scale of 1 : 54 to show yards and feet and long enough to measure 9 yards.
10. A 3.2 cm long line represents a distance of 4 m. Extend the line to measure up to 20 m and show on it units of meter and 5 m. Show a length of 18 m on this scale.

DIAGONAL SCALES

1. Construct a scale of 1 : 40 to read meters , decimeters and centimeters and long enough to measure up to 6 m . Mark a distance of 4.76 m on it.
2. If 1 cm long line on a map represents a real length of 4 m. Calculate the R.F and draw diagonal scale long enough to measure up to 50 meters. Show a distance of 44.5 m on it.
3. The distance between two stations by road is 200 km and it is represented on a certain map by a 5 cm long line. Find the R.F and construct a diagonal scale showing single kilometer and long enough to measure up to 600 km. Show a distance of 467 km on this scale.
4. The distance between two points on a map is 15 cm. The real distance between them is 20 km. Draw a diagonal scale to measure up to 25 km and show a distance of 18.6 km on it.
5. A length of 1 mm is enlarged 20 times in a drawing. What is the R.F of this scale? Draw a diagonal scale of 0.01 mm least count and mark a distance of 7.83 mm on it.
6. The distance between two stations is 100 km and on a road map it is shown by 30 cm. Draw a diagonal scale and mark 46.8 km and 32.4 km on it.

7. Construct a scale to measure kilometer, 1/8th of a kilometer and 1/40th of a kilometer in which 1 km is represented by 3 cm. Mark on this scale a distance of 4.825 km.
8. An area of 400 square centimeters on a map represents an area of 25 square kilometers on a field. Construct a scale to measure up to 5 km and capable to show a distance of 3.56 km. Indicate this distance on the scale.
9. A rectangular field of 0.54 hectares is represented on a map by a rectangle of 3 cm X 2 cm. Calculate the R.F. Draw a diagonal scale to read up to a single meter and long enough to measure up to 500 meters. Show a distance of 438 m on it.
10. A room of 1728 cubic meters volume is shown by a cube of 4 cm side. Find the R.F and construct a scale to measure up to 50 m. Indicate a distance of 37.6 m on the scale.
11. Construct a diagonal scale of 1 : 63360 to read miles, furlongs and chains and long enough to measure up to 6 miles. Show a distance of 4 miles 3 furlongs 2 chains on it.
12. Construct a diagonal scale showing yards, feet and inches in which 2 inches long line represents 1.25 yards and is long enough to measure up to 5 yards. Find R.F and mark a distance of 4 yards 2 feet 8 inches.
13. A rectangle field of 25000 sq. m is represented on a map by a rectangle of 5 cm X 4 cm side. Calculate and long enough to measure up to 500 m. Mark a length of 362 m on the scale.
14. Construct a diagonal scale of R.F 1 : 4000 and having a least count of 1 m. Using this scale, construct a right – angled triangle of sides 301 m and 402 m and find the length of hypotenuse.

CONIC SECTIONS

1. Draw the ellipse when the distance of its focus from its directrix is equal to 60 mm and eccentricity is 3/5. Also draw a tangent and a normal to the ellipse at a point 100 mm away from the directrix.
2. Draw an ellipse when the distance of its vertex from its directrix is 24 mm and distance of its focus from directrix is 42 mm.
3. A fixed point is 90 mm from a fixed straight line. Draw the locus of a point P moving in such a way that its distance from the fixed straight line is twice its distance from the fixed point. Name the curve. Draw a tangent and a normal at point 40 mm away from the fixed point.
4. Construct a parabola whose focus is at a distance of 40 mm from the directrix. Draw a tangent and a normal to the parabola at appoint 50 mm away from the principal axis.
5. Construct a parabola using eccentric method whose vertex is at a distance of 30 mm from the focus. Draw a pair of tangents from a point P, outside the curve, 20 mm from the vertex and 40 mm from the focus.
6. Draw the locus of a point which moves in such a manner that its distance from a fixed point is equal to its distance from a fixed straight line. Consider the distance between the fixed point and the fixed line as 60 mm. Name the curve.
7. The focus of a hyperbola is 60 mm from its directrix. Draw the curve when eccentricity is 5/3. Draw a tangent and a normal to the curve at a point 45 mm from the directrix.

8. Draw the hyperbola when the focus and the vertex are 25 mm apart. Consider eccentricity as 3/2. Draw a tangent and a normal to the curve at a point 35 mm from the focus.
9. Draw the hyperbola when its vertex and its focus are at a distance of 15 mm and 40 mm respectively from the directrix. Plot at least six points.
10. A fixed point is 90 mm from a fixed straight line. Draw the locus of a point P moving in such a way that its distance from the fixed point is twice its distance from the fixed straight line. Name the curve.

CYCLOIDS

1. A circle of diameter 50 mm rolls for a distance of 180 mm along a straight line. Draw the locus of a point lying on the circumference of the circle. Name the curve traced.
2. A circular wheel of diameter 40 mm rolls without slipping along a straight line. Draw the curve traced by a point P, lying on the rim for 1.25 revolutions of the wheel. Name the curve traced. Also, draw a tangent and a normal at a point P, when the wheel has travelled 100 mm from its starting position.
3. A circle of diameter 60 mm rolls over a straight line. Draw the locus of a point lying on the circumference of the circle, initially at the topmost position and occupies the bottom most position after the revolution. Name the curve traced. Also, draw a tangent and a normal to the curve at a point 45 mm above the base line.
4. A circle of diameter 50 mm rolls a straight line without slipping. In the initial position the diameter AB of the circle is parallel to the line on which it rolls. Draw the loci of the point A and B for one revolution of the circle.
5. ABC is an equilateral triangle of side 50 mm. Trace the loci of vertices A, B and C when the circle circumscribing triangle ABC rolls without slipping along a fixed straight line for on revolution.
6. A circle of diameter 50 mm rolls on a horizontal line for a half revolution and then on a vertical line downwards for another half revolution. Draw the curve traced out by a point lying on the circumference of the circle, taking the bottom – most point of the circle as the initial position.
7. A circle of diameter 50 mm rolls on a horizontal line for half revolution and then on an inclined line 60^0 with horizontal for another half. Draw the curve traced out by a point P, lying on the circumference of the circle, taking the topmost point of the rolling circle as the initial position of point P.
8. A circle of diameter 50 mm rolls on a horizontal line for a half revolution and then on a vertical line upwards for another half revolution. Draw the curve traced out by a point lying on the circumference of the circle.

EPICYCLOIDS

1. Draw an epicycloid generated by a point on the circumference of a circle of diameter 50 mm which rolls outside another circle of diameter 150 mm for one

revolution. Also, draw a tangent and a normal to the curve, at a point 115 mm from the center of the directing circle.

2. A 40 mm diameter circle rolls outside an arc of radius 70 mm for a circular distance of 120 mm. Trace the path of a point lying on the circumference of the rolling circle, which is in contact with the arc in its initial position. Name the curve.
3. The angle subtended by a directing circle when a circle of diameter 40 mm rolls outside it for one complete revolution is 108^0. Determine the diameter of the directing circle. Trace the path of a point lying on the circumference of the rolling circle.
4. A circle of diameter 60 mm rolls over a circular arc of diameter 200 mm for one-half revolution and has an external contact. Draw the locus of a point lying on the circumference of the circle and initially farthest away from the arc. Name the curve traced. Also, draw a tangent and a normal to the curve at a point 45 mm above the base line.
5. Draw an epicycloid, when the diameters of the rolling and the directing circles are 50 mm. Suggest an alternative name for this curve.
6. Draw the locus of a point lying on the circumference of a wheel of diameter 60 cm for one revolution, when it passes over a segmental arched culvert of radius 1 m. Take a scale of 1:10
7. A rolling circle of diameter AB = 40 mm, rolls on a fixed disc of diameter 60 mm with external contact. Draw the loci of path traced by the points A and B of the rolling circle for one revolution when one of the end points of diameter AB is in contact to the disc at the starting position.
8. Construct a cardioid taking the diameters of both the rolling and directing circles as 60 mm.

HYPOCYLOIDS

1. Draw a hypocycloid of a circle of diameter 50 mm, which rolls inside a circular arc of radius 85 mm for one revolution. Also, draw a tangent and a normal to the epicycloid at a point 50 mm from the centre of the directing circle.
2. A circus man rides on a motor cycle inside a globe of diameter 3.5 m. The motor cycle wheel is 1 m in diameter. Draw the locus of a point lying on the circumference of the wheel of motor cycle for one turn. Take a scale of 1:20.
3. Draw the locus of a point lying on the circumference of a circle of diameter 70 mm, which rolls inside a circle of diameter 140 mm for one rotation.
4. Construct a hypocycloid taking the diameter of the generating circle and radius of directing circle as 60 mm
5. A hypocycloid is in the form of a 120 mm long straight line. Determine the diameters of the rolling and the directing circles. Also, draw the curve.

UNIT – II

PROJECTIONS OF POINTS

1. Draw the front and the top views of a point A, lying 70 mm above the H.P. and 50 mm in front of the V.P. Also draw the side view.
2. Draw the projections of a point B, lying 70 mm above the H.P. and 50 mm behind the V.P.
3. Draw the projections of a point C, lying 70 mm below the H.P. and 50 mm behind the V.P.
4. Draw the projections of a point D, lying 70 mm below the H.P. and 50 mm in front of the V.P.
5. Draw the projections of a point E, lying on the H.P. and 50 mm in front of the V.P.
6. Draw the projections of a point F, lying 70 mm above the H.P. and on the V.P.
7. Draw the projections of a point G, lying on the H.P. and 50 mm behind the V.P.
8. Draw the projections of a point H, lying 70 mm above the H.P. and on the V.P.
9. Draw the projections of a point J, lying on both the H.P. and the V.P.
10. Draw the projections of the following points on a common reference line keeping the distance between their projectors 30 mm apart.
    (a) Point A is 20 mm below the H.P. and 50 mm in front of the V.P.
    (b) Point B is in the H.P. and 40 mm behind the V.P.
    (c) Point C is 30 mm in front of the V.P. and in the H.P.
    (d) Point D is 50 mm above the H.P. and 30 mm behind the V.P.
    (e) Point E is 20 mm below the H.P. and 50 mm behind the V.P.
    (f) Point F is in the V.P. and 50 mm below the H.P
11. A point is 30 mm from the H.P. and 50 mm from the V.P. Draw its projections keeping it in all possible positions.
12. Draw the projections of the following points on a common reference line keeping the distance between their projectors 25 mm apart.
    (a) Point A is 40 mm above the H.P. and 25 mm in front of the V.P.
    (b) Point B is 40 mm above the H.P. and in the V.P.
    (c) Point C is 25 mm in front of the V.P. and in the H.P.
    (d) Point D is 25 mm above the H.P. and 30 mm behind the V.P.
    (e) Point E is in the H.P. and 30 mm behind the V.P.
    (f) Point F is 40 mm below the H.P. and 30 mm behind the V.P.
    (g) Point G is 25 mm below the H.P. and 40 mm in front of the V.P.
    (h) Point H is in the V.P. and 30 mm below the H.P.
13. Draw the projections of the following points on a common reference line keeping the distance between their projectors 30 mm apart.
    (a) Point P is 35 mm below the H.P. and in the V.P.
    (b) Point Q is 40 mm in front of the V.P. and 25 mm below the H.P.
    (c) Point R is 45 mm above the H.P. and 20 mm behind the V.P.
    (d) Point S is 30 mm below the H.P. and 45 mm behind the V.P.
    (e) Point T is both in H.P. and V.P

PROJECTIONS OF STRAIGHT LINES

1. A 50 mm long line PQ is parallel to both the H.P. and the V.P. It is 25 mm in front of the V.P. and 60 mm above the H.P. Draw its projections and determine the traces.
2. A 60 mm long line PQ has its end P 20 mm above H.P. The line is perpendicular to the H.P. and 40 mm in front of the V.P. Draw its projections and locate the traces.
3. A 60 mm long line PQ has its end P 20 mm in front of the V.P. The line is perpendicular to the V.P. and 40 mm above the H.P. Draw the projections of the line and determine its traces.
4. An 80 mm long line PQ has end P 20 mm above H.P. and 40 mm in front of the V.P. The line is inclined at 30° to the H.P. and is parallel to the V.P. Draw the projections of the line and determine its traces.
5. An 80 mm long line PQ is inclined at 30° to the V.P. and is parallel to the H.P. The end P of the line is 20 mm above the H.P. and 40 mm in front of the V.P. Draw the projections of the line and determine its traces.
6. A 60 mm long line PQ lying on the H.P. is inclined at 30° to the V.P. Its end P is 20 mm in front of the V.P. Draw the projections of the line and determine its traces.
7. Draw the projections of a 70 mm long line PQ, situated in the V.P. and inclined at 30° to the H.P. The end P of the line is 25 mm above the H.P. Also, determine the traces of the line.
8. Draw the projections of a 60 mm long line PQ, which is situated both on the H.P. and the V.P. Also, determine the traces of the line.
9. A 50 mm long line AB has its end A, 30 mm above the H.P. and 20 mm in front of the V.P. The front view of the line is a point. Draw its projections.
10. A 60 mm long line AB is parallel to and 20 mm in front of the V.P. The ends A and B of the line are 10 mm and 50 mm above the H.P., respectively. Draw the projections of the line and determine its inclination with the H.P. Also, locate the traces of the line.
11. An 80 mm long line MN has its end M 15 mm in front of the V.P. The distance between the ends projector is 50 mm. The front view is parallel to and 20 mm above reference line. Draw the projections of the line and determine its inclination with the V.P. Also, locate the traces.
12. The top view of a line measures 60 mm. The line is parallel to the V.P. and inclined at 45° to the H.P. One end of the line is 25 mm in front of the V.P. and lies on the H.P. Draw its projections and determine the true length.
13. A 70 mm long line PQ does not have H.T. and V.T. One end of the line is 30 mm in front of the V.P. and 20 mm above the H.P. Draw its projections.
14. A 60 mm long line PQ is inclined at 45° to the V.P. The V.T. is 25 mm above reference line and the H.T. does not exist. Draw the projections of the line when one end of the line is 10 mm in front of the V.P.
15. A 60 mm long line AB is parallel to both the H.P. and the V.P. It is 20 mm above the H.P. and 30 mm in front of the V.P. Draw its projections.
16. A 60 mm long line PQ has its end A 25 mm above the H.P. and 40 mm in front of the V.P. Draw the projections of the line when it is parallel to both the reference planes.
17. A 55 mm long line PQ is perpendicular to the H.P. and 25 mm in front of the V.P. Draw its projections when one end of the line is 15 mm above the H.P.

18. A 50 mm long line AB is perpendicular to the V.P. and 40 mm above the H.P. One end of the line is 10 mm in front of the V.P. Draw its projections and locate the traces.
19. A 70 mm long line PQ is perpendicular to the H.P. and 30 mm in front of the V.P. End P of the line is on the H.P. Draw its projections and locate the traces.
20. A 60 mm long line lying in the V.P. is perpendicular to the H.P. Draw its projections and locate the traces when one end of the line is 15 mm above the H.P.
21. A 50 mm long line situated in the V.P. is parallel to the H.P. The line is at a distance of 40 mm from the H.P. Draw its projections and locate the traces.
22. A 70 mm long line is parallel to and 50 mm in front of the V.P. Draw its projections when the line is situated in the H.P.
23. A 70 mm long line lies both on the H.P. and the V.P. Draw its projections.
24. A 70 mm long line MN is inclined at 45° to the H.P. and parallel to the V.P. The end M is 15 mm above the H.P. and 25 mm in front of the V.P. Draw its projections and locate the traces.
25. An 80 mm long line is inclined at 60° to the V.P. and parallel to the H.P. One end of the line is 30 mm above the H.P. and 10 mm in front of the V.P. Draw its projections and locate the traces.
26. A 70 mm long line CD lying in the V.P. is inclined at 60° to the H.P. One end of the line 15 mm above the H.P. Draw its projections and locate the traces.
27. A 70 mm long line PQ lying in the H.P. is inclined at 45° to the V.P. End P of the line is in the V.P. Draw its projections and locate the traces.
28. A 70 mm long line has an end 40 mm above the H.P. and 20 mm in front of the V.P. The front view of the line is a point. Draw its projections and determine true inclination with the reference planes.
29. Two points P and Q lying in the V.P. are 90 mm apart. The horizontal distance between the points is 60 mm. If the point P is 15 mm above the H.P., find the height of the point Q above the H.P. and the inclination of the line joining P and Q with the H.P.
30. An 80 mm long line is parallel to and 20 mm above the H.P. One end of the line is in the V.P. whereas the other end is 40 mm in front of the V.P. Draw its projections and determine the true inclination of the line with the V.P.
31. A 75 mm long line is parallel to and 40 mm in front of the V.P. The ends of the line are 25 mm and 50 mm above the H.P. Draw its projections and determine the true inclination of the line with the H.P.
32. A 70 mm long line AB is parallel to the V.P. and inclined to the H.P. The end A is 20 mm above the H.P. and 30 mm in front of the V.P. The top view of the line measures 45 mm. Draw its projections.
33. An 80 mm long line has an end 15 mm above the H.P. and 25 mm in front of the V.P. The line is parallel to the V.P. and its top view measures 40 mm. Draw the projections of the line and find its inclination with the H.P.
34. The front view of an 80 mm long line PQ measures 50 mm. The line lies in the H.P. such that one end is 30 mm in front of the V.P. Draw the projections of the line and find its inclination with the V.P.
35. The top view of an 80 mm long line AB measures 55 mm. The line is in the V.P. and its one end being 20 mm above the H.P. Draw its projections and find inclination with the H.P.
36. A line PQ is parallel to the V.P. and inclined at 30° to the H.P. End P is 20 mm from both the reference planes and the top view measures 70 mm. Draw the projections of the line and determine its true length.

37. The front view of a line PQ parallel to the V.P. and inclined 60° to the H.P. is 50 mm. one end of the line is 20 mm in front of the V.P. and 25 mm above the H.P. Draw its projections and determine true length of the line.
38. A 65 mm long line PQ has end P 20 mm in front of the V.P. and 30 mm above the H.P. Draw its projections such that the line does not have traces.
39. An 80 mm long line PQ has an end 5 mm in front of the V.P. The V.T. of the line is 30 mm above the H.P. and the front view of the line is a point. Draw its projections.
40. The front view a line perpendicular to the H.P. measures 70 mm. The H.T. of the line is 25 mm below the reference line. Draw its projections.
41. A 70 mm long line is inclined at 30° to the H.P. The H.T. of the line lies 15 mm below the reference line and V.T. of the line does not exist. Draw its projections when one end of the line is 25 mm above the H.P.
42. A 75 mm long line is inclined at 45° to the V.P. Its V.T. is 30 mm above the reference plane and H.T. does not exist. One end of the line is 20 mm in front of the V.P. Draw its projections.
43. The front view of a line inclined at 60° to the V.P. measures 60 mm. One end of the line is 20 mm in front of the V.P. The V.T. is 15 mm above the reference line and H.T. does not exist. Draw its projections.
44. An electric switch and a bulb fixed on a wall are 5 m apart. The distance between them measured parallel to the floor is 4 metres. If the switch is 1 m above the floor, find the height of the bulb and inclination of line joining the two with the floor

LINE INCLINED TO BOTH H.P AND V.P

1. A 70 mm long line PQ, has its end P 20 mm above the H.P. and 30 mm in front of the V.P. The line is inclined at 45° to the H.P. and 30° to the V.P. Draw its projections.
2. A straight line PQ has its end P 20 mm above the H.P. and 30 mm in front of the V.P. and the end Q is 80 mm above the H.P. and 70 mm in front of the V.P. If the end projectors are 60 mm apart, draw the projections of the line. Determine its true length and true inclinations with the reference planes.
3. A straight line PQ has its end P 20 mm above the H.P. and 30 mm in front of the V.P. and the end Q is 80 mm above the H.P. and 70 mm in front of the V.P. If the end projectors are 60 mm apart, draw the projections of the line. Determine its true length and true inclinations with the reference planes by trapezoid method.
4. A 100 mm long line PQ has its end P 10 mm above the H.P. and 70 mm in front of the V.P. The line is inclined at 60° to the H.P. and 30° to the V.P. Draw its projections.
5. An 80 mm long line AB is inclined at 30° to the H.P. and 45° to the V.P. The end A is 20 mm above the H.P. and lying in the V.P. Draw the projections of the line
6. A 100 mm long line PQ is inclined at 30° to the H.P. and 45° to the V.P. Its mid-point is 35 above the H.P. and 50 mm in front of the V.P. Draw its projections.
7. A 75 mm long line PQ lying in the first angle has its end P on the H.P. and end Q in the V.P. The line is inclined at 45° to the H.P. and 30° to the V.P. Draw its projections.
8. A 70 mm long line PQ has its end P 20 mm above the H.P. and 40 mm in front of the V.P. The other end Q is 60 mm above the H.P. and 10 mm in front of the V.P.

Draw the projections of PQ and determine its inclinations with the reference planes.

9. The top view of 75 mm long line PQ measures 50 mm. The end P is 15 mm above the H.P. and 50 mm in front of the V.P. The end Q is 20 mm in front of the V.P. and above the H.P. Draw the projections of PQ and determine its inclinations with the reference planes.
10. The front and top views of 75 mm long line PQ measures 50 mm and 60 mm, respectively. If the end P of the line is 35 mm above the H.P. and 15 in front of the V.P., draw its projections and locate the traces. Determine the true inclinations of the line PQ with the H.P. and the V.P.
11. The front and top views of a straight line PQ measures 50 mm and 65 mm, respectively. The point P is on the H.P. and 20 mm in front of the V.P. The front view of the line is inclined at 45° to the reference line. Determine the true length of PQ and its true inclinations with the reference planes. Also, locate the trace.
12. A 70 mm long line PQ is inclined at 45° to the V.P. Its end P lies on the H.P. and 15 mm in front of the V.P. The top view of the line measures 60 mm. Draw the projections of the line PQ and determine its inclination with the H.P
13. A 70 mm long line PQ is inclined at 30° to the H.P. The end P is 15 mm in front of the V.P. and 25 mm above the H.P. The front view of the line measures 45 mm. Draw the projections of the line PQ and determine its true angle of inclination with the V.P.
14. The front and top views of an 80 mm long line PQ measures 70 mm and 60 mm, respectively. The end P is on the H.P. and the end Q is in the V.P. Draw the projections of line PQ and determine its inclinations with the H.P. and the V.P. Also, locate the traces.
15. The distance between the end projectors of a line PQ is 50 mm. The end P is 50 mm in front of the V.P. and 25 mm above the H.P. The end Q is 10 mm in front of the V.P. and above the H.P. The line is inclined at 30° to the V.P. Draw the projections of line PQ. Determine its true length and true angle of inclination with the H.P.
16. An 80 mm long line PQ has its end P on the H.P. and 15 mm in front of the V.P. The line is inclined at 30° to the H.P. and its top view is inclined at 60° to the reference line. Draw the projections of line PQ and determine true angle of inclination with the V.P.
17. An 80 mm long line PQ has its end P 10 mm above the H.P. and 25 mm in front of the V.P. The line is inclined at 30° to the H.P. and 60° to the V.P. Draw its projections.
18. A 75 mm long line PQ has its end P 15 mm above the H.P. and 20 mm in front of the V.P. The front and top views are 45 mm and 60 mm long, respectively. Draw the projections of line PQ and determine its true inclinations with the reference planes.
19. An 80 mm long line PQ has its end P 15 mm above the H.P. and 50 mm in front of the V.P. while the end Q is in the V.P. Draw the projections of the line when the sum of its inclination with the H.P. and V.P. is 90°. Determine the true inclinations of line PQ with the reference planes and locate its traces.
20. Two straight lines PQ and QR make an angle of 120° between them in their front and top views. PQ is 60 mm long and is parallel to and 15 mm from both the H.P. and the V.P. Determine the true angle between PQ and QR, if point R is 50 mm above H.P.

21. Find graphically the length of the largest rod that can be kept inside a hollow cuboid of 60 mm x 40 mm x 30 mm.
22. A room is 6 m x 5 m x 4 m high. An electric bulb B is above the centre of the longer wall and 1 m below the ceiling. The bulb B is 50 cm away from the longer wall. The switch S for the light is 1.25 m above the floor on the centre of the adjacent wall. Determine graphically, the shortest distance between the bulb B and the switch S.
23. Three wires AB, CD and EF are tied at points A, C, E on a vertical pole 14 m long at height 12 m, 10 m and 8 m respectively from the ground. The lower ends of the wires are tied to hooks at point B, D and F on the ground level all of which lie at the corners of an equilateral triangle of 7.5 m side. If the pole is situated at the centre of the triangle, determine the length of each rope and their inclination with the ground. (Use scale 1:100).
24. An 80 mm long line PQ is inclined at 45° to the H.P. and 30° to the V.P. The end P is in the H.P. and 40 mm in front of the V.P. Draw its projections.
25. An 80 mm long line PQ has the end Q lying both in the H.P. and V.P. The line is inclined at 30° to H.P. and 45° to the V.P. Draw its projections.
26. A 100 mm long line PQ is inclined at 45° to H.P. and 30° to the V.P. Its mid-point is 50 above the H.P. and 40 mm in front of the V.P. Draw its projections.
27. A 120 mm long line PQ is inclined at 45° to the H.P. and 30° to the V.P. A point M lies on PQ at a distance of 40 mm from the end P. The front and top views of point M is 40 mm above xy and the top view is 35 mm below xy. Draw the projections of PQ.
28. A 100 mm long straight line PQ lying in the first angle. It is inclined at 30° to the H.P. and 45° to the V.P. The point R divides the line in the ratio of 1:3 and is situated 25 mm above the H.P. and 30 mm in front of the V.P. Draw the projections of line PQ.
29. An 80 mm long line PQ lying in the first angle, has the end P on the H.P. and end Q in the V.P. The line is inclined at 45° to the H.P. and 30° to the V.P. Draw its projections.
30. A straight line PQ has its end P 15 mm above the H.P. and 60 mm in front of the V.P. The end Q is 45 mm above the H.P. and 10 mm in front of the V.P. If the end projectors of the line are 60 mm apart, draw its projections. Determine the true length and true inclinations of the line with the principal planes.
31. A 90 mm long line PQ has the end P 20 mm above the H.P. and 35 mm in front of the V.P. The end Q is 80 mm above the H.P. and 60 mm in front of the V.P. Draw the projections of PQ and determine its true inclinations with the principal planes.
32. A line PQ has its end projectors 50 mm apart. The end P is 20 mm above the H.P. and 15 mm in front of the V.P. while the end Q is 60 mm above the H.P. and 70 mm in front of the V.P. Draw the projections of the line and determine its true length and inclinations with the principal planes. Also, locate its traces.
33. A 75 mm long line PQ is inclined at 30° to the H.P. Its end P is 20 mm above the H.P. and on the V.P. End Q is 60 mm in front of the V.P. Draw the projections of the line and locate its traces.
34. A line PQ, 75 mm long is inclined at an angle of 30° to the V.P. The end P is on the H.P. and 30 mm in front of the V.P. The end Q is 50 mm above the H.P. Draw the projections of the line and locate its traces.
35. A 75 mm long line PQ is inclined at 45° to the H.P. The end P is 15 mm above the H.P. and 25 mm in front of the V.P. A vertical plane containing line PQ is

inclined at 45° to the V.P. Draw the projections of the line and determine its inclination with V.P. Also locate the traces.

36. A 75 mm long line PQ has the end P 25 mm above the H.P. and 10 mm in front of the V.P. The length of the front and top views are 60 mm. Draw the projections of the line and locate its traces.
37. A line PQ has its end P 10 mm above the H.P. and 15 mm in front of the V.P. The lengths of its front and top views are 50 mm and 60 mm respectively. If the top view of the line is inclined at 30° to the reference line, draw its projections. Determine the true length and true inclinations of the line with the principal planes. Also, locate the traces.
38. A line PQ inclined at 30° to the H.P. has the end P 20 mm above the H.P. and 10 mm in front of the V.P. The front view of the line measures 70 mm and is inclined at 60° to the reference line. Draw the projections of the line and determine its true length and inclinations with the principal planes. Also, locate its traces.
39. A line PQ inclined at 45° to the V.P. has its front view 60 mm long. The end P is 10 mm from both the principal planes while the end Q is 45 mm above the H.P. Draw the projections of the line and determine its true length and inclinations with the principal planes. Also, locate the traces.
40. A line PQ has the end projectors 50 mm apart. The front and top views of the line are 55 mm and 65 mm long respectively. If the end P is 15 mm above the H.P. and 25 mm in front of the V.P., draw the projections of the line. Determine its true length and inclinations with the principal planes. Also, locate the traces.
41. A line PQ inclined at 45° to the H.P. has its top view 60 mm long. The end P is 15 mm above the H.P. and 20 mm in front of the V.P. whereas the end Q is equidistant from both the principal planes. Draw the projections of the line and determine its true length and inclination with the V.P. Also, locate its traces.
42. A 120 mm long line PQ has the ends P and Q, 10 mm and 60 mm below the H.P. respectively. The end projectors are 50 mm apart. The mid-point of PQ is 60 mm in front of the V.P. Draw the projections of PQ and determine its inclinations with the reference planes.
43. A line PQ is inclined at 30° to the H.P. The end P is 15 mm in front of the V.P. and the mid-point O of the line is 40 mm above the H.P. The front view measures 60 mm and is inclined at 45° with the reference line. Draw the projections of the line and determine its true length and inclination with V.P. Also, locate its traces.
44. An 80 mm long line PQ is inclined at 30° to the H.P. and 45° to the V.P. End P is on the H.P. and end Q is in the V.P. Draw projections of the line and locate its traces.
45. A 90 mm long line PQ is inclined at 30° to the H.P. and 45° to the V.P. The end P is 10 mm above the H.P. and 100 mm in front of the V.P. Draw the projections of the line and determine distance of end P from the V.P. Also, locate the traces.
46. An 80 mm long line PQ is inclined at 30° to the H.P. and 45° to the V.P. The end P is 15 mm in front of the V.P. and the end Q is 80 mm above the H.P. Draw the projections of the line and determine distance of end P from the H.P. Locate its traces.
47. A line PQ is inclined at 30° to the H.P. and 45° to the V.P. The end P is 30 mm above the H.P. and end Q is 70 mm in front of the V.P. If the top view of the line measures 75 mm, draw its projections. Determine its true length, inclination with the V.P. and the traces.

48. An 80 mm long line PQ is inclined at 30° to the H.P. and 45° to the V.P. The end P is 15 mm above the H.P. and the end Q is 15 mm in front of the V.P. Draw projections of the line and locate its traces.

49. A 90 mm long line PQ has its end P 15 mm above the H.P. and 25 mm in front of the V.P. The line is inclined at 60° to the H.P. and 30° to the V.P. Draw its projections.

50. A 100 mm long line PQ has its end P 15 mm above the H.P. and 20 mm in front of the V.P. The front and top views are 80 mm and 60 mm long, respectively. Draw its projections and determine the true inclinations with the reference planes.

51. A 90 mm long line PQ has the end P on the H.P. and 70 mm in front of the V.P. The end Q is 10 mm in front of the V.P. Draw the projections of the line when the sum of its true inclinations with the H.P. and the V.P. is 90°. Determine the true inclinations with the reference planes and locate its traces.

PROJECTIONS OF PLANES

1. A square plane of side 40 mm has its surface parallel to and 20 mm above the H.P. Draw its projections when (a) a side is parallel to the V.P., (b) one side is inclined at 30° to the V.P. and (c) all sides are equally inclined to the V.P.
2. A hexagonal plane of side 25 mm has its surface parallel to and 20 mm in front of V.P. Draw its projections, when a side is (a) parallel to the H.P., (b) perpendicular to the H.P., (c) inclined at 45° to the H.P.
3. A triangular plane is in the form of an isosceles triangle of base side 30 mm and altitude 40 mm. Its surface is perpendicular to both H.P. and V.P. Draw its projections when the base side is parallel to the V.P.
4. A hexagonal plane of side 30 mm has an edge on the H.P. The surface is inclined at 45° to the H.P. and perpendicular to the V.P. Draw its projections.
5. A hexagonal plane of side 30 mm has a corner on the ground. Its surface is inclined at 45° to the H.P. and perpendicular to the V.P. Draw its projections when the diagonal through the corner in the H.P. is parallel to the V.P.
6. A circular plane of diameter 50 mm is resting on a point of the circumference on the H.P. The plane is inclined at 30° to the H.P. and its centre is 35 mm in front of the V.P. Draw its projections.
7. A rectangular plane of sides 70 mm and 35 mm has a shorter side on the H.P. The surface of the plane is inclined at 60° to the H.P. and perpendicular to the V.P. Draw its projections.
8. A rectangular plane of sides 70 mm and 35 mm is resting on a side on the H.P. The surface is inclined to the H.P. and perpendicular to the V.P. such that the top view appears as a square. Draw its projections and determine inclination of the plane with the H.P.
9. A rhombus of diagonals 70 mm and 45 mm is placed on an end of the major diagonal on the H.P. Its surface is inclined at 50° to the H.P. and perpendicular to the V.P. Draw its projections.
10. A rhombus of diagonals 70 mm and 45 mm is placed on an end of the major diagonal on the H.P. Its surface is inclined to the H.P. and perpendicular to the V.P. such that the top view appears as a square. Draw its projections and determine inclination of the rhombus with the H.P.
11. The top view of a plane whose surface is perpendicular to the V.P. and inclined at 45° to the H.P. is a circle of diameter 50 mm. Draw the projections of the plane and determine its true shape.
12. The top view of a plane is a regular pentagon of side 30 mm having one side inclined at 30° to the V.P. Its front view is a straight line inclined at 45° to the reference line. Draw the projections of the plane and determine its true shape.
13. A hexagonal plane of side 30 mm has an edge in the V.P. The surface of the plane is inclined at 45° to the V.P. and perpendicular to the H.P. Draw its projections.
14. A hexagonal plane of side 30 mm has a corner in the V.P. The surface of the plane is inclined at 45° to the V.P. and perpendicular to the H.P. Draw its projections. Assume that the diagonal through the corner in the V.P. is parallel to the H.P.
15. A circular plane of diameter 50 mm is resting on a point of the circumference on the V.P. The plane is inclined at 30° to the V.P. and the centre is 35 mm above the H.P. Draw its projections.

16. An isosceles triangle of base 40 mm and altitude 54 mm has its base in the V.P. The surface of the plane is inclined at 50° to the V.P. and perpendicular to the H.P. Draw its projections.
17. An isosceles triangle of base 40 mm and altitude 54 mm is placed in such a way that it appears as an equilateral triangle in the front view. Draw its projections and determine inclination of the triangle with the V.P.
18. A 60° set-square has the shortest edge of 40 mm lying in the V.P. The surface is inclined to the V.P. and perpendicular to the H.P. such that the front view appears as an isosceles triangle. Draw the projections of the set-square and determine its inclination with the V.P.
19. A square lamina ABCD of side 40 mm is suspended from a point O such that its surface is inclined at 30° to the V.P. The point O lies on the side AB 12 mm away from A. Draw its projections.
20. A rectangular plane of sides 40 mm and 60 mm has a corner on the H.P. and 20 mm in front of the V.P. The surface of the plane is parallel to the V.P. and all the sides are equally inclined to the H.P. Draw its projections and locate the traces.
21. A thin hexagonal plate of side 30 mm has a side inclined at 45° to the V.P. Its V.T. is parallel to and 25 mm above the reference line. Draw its projections.
22. A hexagonal plane of side 25 mm has its surface parallel to and 20 mm above the H.P. Draw its projections, when a side is (a) parallel to V.P., (b) perpendicular to V.P., (c) inclined at 45° to V.P.
23. A circular plane of diameter 60 mm has its centre 20 mm above the H.P. and 35 mm in front of the V.P. The surface of the plane is parallel to the H.P. Draw its projections.
24. A composite plate of negligible thickness is made up of a rectangle with sides 60 mm and 40 mm and a semicircle on its longer side. The plate lies in the H.P. with one of the shorter sides parallel to the V.P. Draw its projections.
25. A pentagonal plane of side 35 mm has a corner on the H.P. and the side opposite to this corner is parallel to the H.P. The plane is parallel to and 20 mm in front of the V.P. Draw its projections and locate its traces.
26. A square ABCD of side 40 mm is suspended from a point O, lying on side AB at a distance 15 mm from corner A. The plane is parallel to and 25 mm in front of the V.P. Draw its projections and locate the traces.
27. A rectangular plane of sides 50 mm and 30 mm is perpendicular to both H.P. and V.P. The longer edges are parallel to the H.P. and nearest one is 20 mm above it. The shorter edge nearer to V.P. is 15 mm from it. Draw its projections.
28. A thin hexagonal plate of side 30 mm has one of the sides inclined at 45° to the VP. Its V.T. is parallel to and 25 mm above the reference line and the H.T. does not exist. Draw its projections.
29. An equilateral triangular plane of sides 60 mm has a side inclined at 45° to the H.P. Its H.T. is parallel to and 25 mm below the reference line. The V.T. does not exist. Draw its projections. Surface inclined to H.P. and perpendicular to V.P.
30. A pentagonal plane of side 30 mm has an edge on the H.P. The surface of the plane is inclined at 45° to the H.P. and perpendicular to the V.P. Draw its projections and locate the traces.
31. A pentagonal plane of side 30 mm has a corner on the ground. The surface of the plane is inclined at 45° to the H.P. and perpendicular to the V.P. Draw its projections when the side opposite to the corner on which it is resting is parallel to H.P.

32. A circular plane of diameter 50 mm is resting on a point of the circumference on the H.P. The plane is inclined at 45° to the H.P. and the centre is 35 mm in front of the V.P. Draw its projections and locate the traces.
33. A trapezium ABCD having parallel sides AB = 50 mm, CD = 30 mm and height 40 mm is kept on its side AB in the V.P. The surface of the trapezium is inclined at 45° to the H.P. and perpendicular to the V.P. Draw its projections.
34. A square lamina of side 50 mm has a corner on the H.P. The diagonal through that corner is parallel to V.P. and inclined at 30° to the H.P. Draw its projections when the lamina is perpendicular to the V.P. Measure the distance of the topmost corner from the H.P.
35. A pentagonal plane of side 30 mm has a side on the H.P. and the surface perpendicular to the V.P. The corner opposite to that side is 40 mm above the H.P. Draw the projections of the plane and determine its inclination with the H.P.
36. A rectangular plate of sides 60 mm and 40 mm rest on a shorter edge on the H.P. with its surface perpendicular to the V.P. The centre of the plate is 20 mm above the H.P. and 30 mm in front of the V.P. Draw the projections of the plate and determine angle made by it with the H.P.
37. A hexagonal plate of side 30 mm has a centrally punched circular hole of diameter 30 mm. The plane is placed on a side in the H.P. such that the surface is perpendicular to the V.P. and inclined at 45° to the H.P. Draw its projections when the centre of the plate is 35 mm in front of the V.P.
38. A square plane of side 50 mm is placed on the H.P. The surface is inclined to the H.P. and perpendicular to the V.P. such that the top view appears as a rectangle of 50 mm and 25 mm sides. Draw its projections and determine inclination of the plane with the H.P.
39. A square plane of diagonal 70 mm is kept in such a way that its top view appears as a rhombus of 70 mm and 45 mm diagonals. Draw its projections and determine inclination of the plane with the H.P.
40. A 60° set-square has shortest edge of 50 mm in the H.P. The surface is inclined to the H.P. and perpendicular to the V.P. such that the top view appears as isosceles triangle. Draw its projections and determine inclination of the set-square with the H.P.
41. A circular plane of diameter 50 mm is placed on a point of the circumference in the H.P. Its surface is inclined to the H.P. and perpendicular to the V.P. such that top view appears as an ellipse of major and minor axes 50 mm and 30 mm, respectively. Draw the projections of the plane and determine its inclination with the H.P.
42. The top view of a plane is a circle of diameter 50 mm. Its front view is a straight line inclined at 45° to the reference line. Draw the projections of the plane and determine its true shape.
43. The top view of a plane whose surface is inclined at 45° to the H.P. and perpendicular to V.P. appears as a regular hexagon of side 30 mm with a side parallel to the reference line. Draw the projections of the plane and determine its true shape.
44. The top view of a plane is a square of side 40 mm having one side inclined at 30° to the V.P. Its front view is a straight line inclined at 45° to the reference line. Draw the projections of the plane and determine its true shape.
45. The top view of a triangular plane is an equilateral triangle of side 50 mm having one side parallel to the reference line. The front view of the plane is a 65

mm long straight line. Determine the true shape of the plane. Surface inclined to V.P. and perpendicular to H.P.

46. A pentagonal plane of side 30 mm has an edge in the V.P. The surface of the plane is inclined at 45° to the V.P. and perpendicular to the H.P. Draw its projections and locate the traces.
47. A pentagonal plane of side 30 mm has a corner in the V.P. The surface of the plane is inclined at 45° to the V.P. and perpendicular to the H.P. Draw its projections when the side opposite to the corner in the V.P. is parallel to the V.P.
48. A hexagonal lamina of side 50 mm has a corner in the V.P. The diagonal through that corner is parallel to H.P. and inclined at 30° to the V.P. Draw its projections when the lamina is perpendicular to the H.P. Measure the distance of the topmost corner from the V.P.
49. A pentagonal plane of side 30 mm rests on an edge in the V.P. with its surface perpendicular to the H.P. The corner opposite to the edge on which it is resting, is 30 mm in front of the V.P. Draw the projections of the plane and determine its inclination with the V.P.
50. A hexagonal plane of side 30 mm has a centrally punched circular hole of 30 mm diameter. An edge of the plane is in the V.P. Its surface is perpendicular to the H.P. and inclined at 45° to the V.P. Draw its projections.
51. A circular lamina of diameter 50 mm has its centre 35 mm above the H.P. and 25 mm in front of the V.P. The surface of lamina is inclined at 45° to the V.P. and perpendicular to the H.P. Draw its projections.
52. A square plane of side 50 mm is placed in the V.P. The surface is inclined to the V.P. and perpendicular to the H.P. such that the front view appears as a rectangle of 50 mm and 25 mm sides. Draw its projections and determine inclination of the plane with the V.P.
53. A square plane of diagonal 60 mm is placed on the V.P. The surface is inclined to the V.P. and perpendicular to the H.P. such that front view appears as a rhombus of 60 mm and 46 mm diagonals. Draw its projections and determine inclination of the plane with the V.P.
54. A rectangular plane of side 35 mm and 70 mm is placed on a side in the V.P. The surface is inclined to the V.P. and perpendicular to the H.P. such that the front view appears as a square. Draw its projections and determine inclination of the plane with the V.P.
55. The major and the minor diagonals of a rhombus are 70 mm and 45 mm respectively. It is placed in such a way that its front view appears as a square. Draw its projections and determine inclination of the rhombus with the V.P.
56. A circular plate of diameter 50 mm is resting on a point of the rim in the V.P. Its surface is inclined to the V.P. and perpendicular to H.P. such that the front view appears as an ellipse of major and minor axes 50 mm and 30 mm, respectively. Draw the projections of the plate and determine its inclination with the V.P.
57. The front view of a plane whose surface is perpendicular to H.P. and inclined at 30° to the V.P. appears as a regular pentagon of side 30 mm with a side parallel to the reference line. Draw the projections of the plane and determine its true shape.
58. The front view of a plane is a straight line inclined at 30° with the reference line. Its top view is a regular hexagon of side 30 mm having an edge inclined at 45° with the reference line. Draw its projections and determine its true shape

UNIT – III

PROJECTIONS OF SOLIDS

AXIS PERPENDICULAR TO H.P

1. A square pyramid of base side 40 mm and axis 60 mm is resting on its base on the H.P. Draw its projections when (a) a side of the base is parallel to the V.P., (b) a side of the base is inclined at 30° to the V.P., (c) all the sides of the base are equally inclined to the V.P.
2. A square prism of base side 40 mm and axis 60 mm is resting on its base on the ground. Draw its projections when (a) a face is perpendicular to the V.P., (b) a face is inclined at 30° to the V.P., (c) all the faces are equally inclined to the V.P.
3. A pentagonal prism of base side 30 mm and axis 60 mm has one of its bases in the V.P. Draw its projections when (a) a rectangular face is parallel to and 15 mm above the H.P., (b) a face is perpendicular to the H.P., (c) a face is inclined at 45° to the H.P
4. A pentagonal prism of base side 30 mm and axis 60 mm is resting on one of its rectangular faces on the H.P. with an axis parallel to the V.P. Draw its projections.
5. A hexagonal prism of base edge 30 mm and axis 70 mm has its axis parallel to and 50 mm above the H.P. Its base is parallel to the V.P. and an edge of the base is inclined at 45° to the H.P. Draw its projections.
6. A hexagonal prism of base side 25 mm and axis 60 mm is resting on one of its rectangular faces on the H.P. with the axis perpendicular to the V.P. A right circular cone of base 40 mm diameter and axis 45 mm is placed centrally on the top of the prism. Draw the projections of the composite solid.
7. A tetrahedron of edge 65 mm has a face in the V.P. and an edge of that face is perpendicular to the H.P. Draw its projections.
8. A tetrahedron of edge 60 mm is resting on a face on the H.P. such that an edge is parallel to and 20 mm in front of the V.P. Draw its projections.

Axis perpendicular to H.P.

9. A tetrahedron of edge 60 mm is resting on a face on the H.P. such that an edge is parallel to and 20 mm in front of the V.P. Draw its projections.
10. Draw the projections of a hexagonal prism, base side 30 mm and axis 70 mm, resting on its base on the H.P. such that a side of the base is (a) parallel to the V.P., (b) perpendicular to the V.P., and (c) inclined at 45° to the V.P.
11. A cone of base diameter 50 mm and axis 60 mm has its base parallel to and 10 mm above the H.P. while the axis is parallel to and 40 mm in front of the V.P. Draw its projections.
12. A square prism of base side 45 mm and axis 65 mm has its axis parallel to and 55 mm in front of V.P. An edge of its base is parallel to the H.P. and inclined at 30° to the V.P. Draw its projections.
13. Draw the projections of a frustum of a cone of base diameter 60 mm, top diameter 30 mm and height 75 mm resting on its base on the H.P.

14. Draw the projections of the frustum of a square pyramid base edge 50 mm, top edge 25 mm and height 60 mm resting on its base on the H.P. with a side of base inclined at 30° to the V.P.
15. A cube of edge 45 mm is resting on the H.P. with a vertical face inclined at 30° to the V.P. Draw its projections.
16. An octahedron of edge 40 mm rests on an apex on the H.P. Its axis perpendicular to H.P. and a horizontal edge is inclined at 30° to the V.P. Draw its projections.
17. A square slab of base side 60 mm and thickness 20 mm is resting on its base on the ground with an edge inclined at 30° to the V.P. A cone of base diameter 50 mm and axis 60 mm is placed centrally over the slab such that axes of the solids coincide. Draw its projections.
18. A cone of base diameter 40 mm and axis 55 mm is resting centrally on a hexagonal slab of side 25 mm and thickness 15 mm. Draw the projections of the arrangement when one of the sides of the base of the slab is inclined at 45° to the V.P.
19. A hexagonal prism of base side 30 mm and axis 50 mm is resting on its base on the H.P. with a side of base perpendicular to the V.P. A tetrahedron is placed on the top of the prism such that the three corners of the tetrahedron coincide with the alternate corners of the prism. Draw the projections of the arrangement.

Axis perpendicular to V.P.

20. Draw the projections of a square prism, base side 40 mm and axis 65 mm, having its base in the V.P. such that (a) a rectangular face is parallel to the H.P., (b) two faces are inclined at 30° to the H.P., and (c) all faces are equally inclined to the H.P.
21. Draw the projections of a hexagonal pyramid, base side 30 mm and axis 60 mm, resting on its base on the V.P. with a side of base (a) parallel to the H.P., (b) perpendicular to the H.P., and (c) inclined at 45° to the H.P.
22. A cylinder of base diameter 50 mm and axis 65 mm has its axis 40 mm above H.P. and perpendicular to V.P. Draw its projections when one of the bases being 10 mm in front of the V.P.
23. A cone of base diameter 50 mm and axis 65 mm has its apex in the V.P. Its axis is parallel to and 40 mm above the H.P. Draw its projections.
24. A tetrahedron of edge 60 mm is resting on one face on the V.P. such that one of the edges is parallel to and 20 mm above H.P. Draw its projections.
25. Draw the projections of a triangular prism of base edge 50 mm and axis 65 mm is placed on one of its rectangular faces on the H.P. such that the axis is perpendicular to the V.P.
26. A cube of edge 40 mm is resting on one of its edges on the H.P. with the faces containing the resting edge equally inclined to H.P. and the two vertical faces parallel to the V.P. Draw its projections.
27. A pentagonal pyramid of base side 25 mm and axis 60 mm rests on an edge of the base on the H.P. with axis perpendicular to V.P. Draw its projections when the base is 15 mm in front of V.P.
28. A circular disc of diameter 60 mm and thickness 20 mm is resting on its base on the H.P. A hexagonal prism of side 25 mm and axis 80 mm is surmounted centrally on the disc such that a face of the prism lies on the top of the disc and

the axis of the prism is perpendicular to the V.P. Draw the projections of the arrangement.

Axis parallel to both H.P. and V.P.

29. A hexagonal prism of base side 30 mm and axis 70 mm, rests on a rectangular face on the H.P. such that the axis is parallel to and 45 mm in front of the V.P. Draw its projections.

Axis Inclined to H.P

1. A pentagonal prism of base edge 30 mm and axis 60 mm rests on an edge of its base in the H.P. Its axis is parallel to V.P. and inclined at 45° to the H.P. Draw its projections.
2. A pentagonal prism of base side 30 mm and axis 70 mm has a corners on the H.P. and the axis is inclined at 45° to the H.P. Draw its projection when the plane containing the resting corner and the axis is parallel to the V.P.
3. A hexagonal pyramid of base side 30 mm and axis 60 mm has an edge of its base on the ground. Its axis is inclined at 30° to the ground and parallel to the V.P. Draw its projections
4. A hexagonal pyramid of base edge 30 mm and axis 60 mm, has a triangular face on the ground and the axis parallel to the V.P. Draw its projections.
5. A hexagonal pyramid of base edge 30 mm and axis 60 mm, is lying on a slant edge on the ground with the axis parallel to the V.P. Draw its projections when the face containing the resting edge are equally inclined to the H.P.
6. A cylinder of base diameter 50 mm and axis 70 mm has a generator in the V.P. and inclined at 45° to the H.P. Draw its projections.
7. A pentagonal pyramid of base side 30 mm and axis 60 mm has an e
8. dge of base parallel to H.P. Its axis is parallel to V.P. and inclined at 45° to the H.P. Draw its projections when the apex lies in the H.P.
9. Draw projections of a cube of edge 45 mm, resting on one of its corners in the H.P. with a solid diagonal vertical.
10. A tetrahedron of edge 70 mm has an edge on the ground and the faces containing that edge are equally inclined to the H.P. Draw its projections when the edge lying on the ground is perpendicular to the V.P.
11. A hexagonal pyramid of base side 25 mm and axis 60 mm is freely suspended from one of the corners of the base. Draw its projections when its axis is parallel to the V.P.
12. A hexagonal prism of base edge 30 mm and axis 70 mm has an edge of its base in the V.P. such that the axis is inclined at 30° to the V.P. and parallel to the H.P. Draw its projections.
13. A pentagonal pyramid of base side 30 mm and axis 55 mm has a triangular face in the V.P. and the base edge contained by that triangular face is perpendicular to the H.P. Draw its projections.
14. A cone of base diameter 50 mm and axis 60 mm has a generator in the V.P. and the axis parallel to the H.P. Draw its projections.
15. A pentagonal prism of base side 30 mm and axis 60 mm has one of its rectangular faces on the H.P. and the axis inclined at 60° to the V.P. Draw its projections.

16. Draw three views of the frustum of a hexagonal pyramid of base edge 35 mm, top edge 20 mm and axis 45 mm, has one of its faces on the H.P. and axis parallel to V.P.
17. A hexagonal prism, base side 40 mm and axis 40 mm has a centrally drilled circular hole of diameter 40 mm. Draw its projections when the prism is resting on an edge of its base on the H.P. and the axis inclined at 60° to the H.P. and parallel to the V.P.
18. A hexagonal pyramid of base side 25 mm and height 50 mm is resting centrally on the flat surface of a cylindrical disc of base diameter 60 mm and thickness 20 mm, such that one of the edges of the base of the pyramid is parallel to the V.P. Draw the projections of the combined solid when their common axis is inclined at 45° to the H.P. and parallel to the V.P.

Axis inclined to H.P. and parallel to V.P.

19. A square prism of base edge 35 mm and axis 60 mm is resting on an edge of its base such that the axis is parallel to V.P. and inclined at 45° to the H.P. Draw its projections.
20. A pentagonal pyramid of base edge 30 mm and axis 60 mm is resting on a corner of its base on the H.P. Draw its projections when its axis is parallel to the V.P. and inclined at 60° to the H.P.
21. A cone of base diameter 50 mm and height 60 mm is resting on a point of its base circle on the H.P. with its axis parallel to the V.P. and inclined at 45° to the H.P. Draw its projections.
22. A pentagonal prism, base side 30 mm and axis 80 mm, stands on a corner of its base on the H.P. with the longer edge through that corner inclined at 45° to the H.P. and the face opposite to this slant edge is perpendicular to the V.P. Draw its projections.
23. A square pyramid of base edge 40 mm and axis 60 mm is resting on a triangular face on the V.P. Draw its projections when the axis is parallel to and 30 mm above the H.P.
24. A hexagonal pyramid of base side 30 mm and axis 60 mm is lying on one of its triangular faces in the H.P. Draw its three views.
25. A cone of base diameter 50 mm and axis 65 mm is lying on one of its generators on the H.P. Draw its projections.
26. Draw three views of a square pyramid of base side 40 mm and axis 60 mm, which is resting on a corner of its base on the H.P. and the slant edge containing that corner is vertical.
27. A cylinder of base diameter 50 mm and axis 70 mm is resting on a point of its base circle on H.P. Draw its projections when its generators are inclined at 45° to the H.P.
28. A pentagonal prism of base side 25 mm and axis 60 mm is resting on one of its sides of the base on the H.P. and the plane of the base is inclined at 60° to the H.P. Draw its projections.
29. A pentagonal pyramid of base side 30 mm and axis 60 mm is resting on a corner of its base on the H.P. such that the highest edge of the base is 25 mm above the H.P. Draw its projections.
30. A pentagonal prism of base side 30 mm and axis 70 mm is resting on a corner of its base on the H.P. such that the highest edge of the prism is parallel to and 80 mm above H.P. Draw its projections.

31. A cone of base diameter 50 mm and axis 70 mm is resting on a point of its base circle on the H.P. and the highest point of the base circle is 25 mm above the H.P. and 40 mm in front of the V.P. Draw its projections.
32. A pentagonal prism, of base side 30 mm and axis 70 mm is resting on one of its rectangular faces in the V.P. with axis inclined at 60° to the H.P. Draw its projections.
33. A frustum of a pentagonal pyramid has base edge 40 mm, top edge 25 mm and height 60 mm is resting on an edge of its base on the H.P. Draw its projections when the axis is inclined at 45° to the H.P. and parallel to the V.P.
34. A cube of edge 40 mm is resting on a corner on the H.P. and one of its solid diagonal is parallel to both the reference planes. Draw its three views.
35. A tetrahedron of side 60 mm is resting on an side on the H.P. such that a face perpendicular to V.P. is inclined at 45° to the H.P. Draw its projections.
36. An octahedron of edge 40 mm is resting on a face on the H.P. and the axis is parallel to the V.P. Draw its projections.
37. A square slab of side 60 mm and thickness 25 mm has a coaxial hole of 40 mm diameter. Draw its projections when it is resting on a corner in the H.P. and the axis is inclined at 30° to the H.P.
38. A pentagonal pyramid of base side 30 mm and height 70 mm is resting axially on the flat surface of a cylindrical block of base diameter 60 mm and thickness 20 mm, such that an edge of the base of the pyramid is parallel to the V.P. Draw the projections of the solids when the axis is inclined at 45° to H.P. and parallel to the V.P

Axis inclined to V.P. and parallel to H.P.

39. A hexagonal pyramid, base side 30 mm and axis 60 mm, rests on an edge of its base on the V.P. with a triangular face containing that edge inclined at 45° to the V.P. and perpendicular to the H.P. Draw its projections.
40. A hexagonal prism of base side 30 mm and axis 70 mm is resting on a rectangular face on the H.P. and a longer edge is making 60° with the V.P. Draw its projections.
41. A cone of base diameter 50 mm and axis 60 mm has its axis parallel to H.P. and inclined at 30° to the V.P. Draw its projections.
42. A triangular prism of base side 40 mm and axis 65 mm is resting on a rectangular face on the H.P. Draw its projections when its axis is inclined at 30° to the V.P.
43. A cylinder of base diameter 50 mm and axis 65 mm is resting on one of its generators on the H.P. Draw its projections when the axis is inclined at 30° to the V.P.
44. A hexagonal pyramid of base side 30 mm and axis 60 mm has an edge of its base in the V.P. and the apex is 50 mm away from both the reference planes. Draw its projections when the axis is parallel to the H.P.
45. A cube of edge 40 mm is resting on a corner on the V.P. and the solid diagonal through that corner is perpendicular to the V.P. Draw its projections.
46. Draw the projections of a tetrahedron of edge 60 mm having an edge in the V.P. and another edge parallel to both the reference planes.

AXIS INCLINED TO BOTH H.P. AND V.P.

An Element of the Solid in the H.P.

1. A square prism of base edge 35 mm and axis 60 mm is resting on an edge of its base on the H.P. and the axis inclined at 45° to the H.P. If the edge resting on the H.P. is inclined at 30° to the V.P., draw its projections.
2. A pentagonal prism of base side 30 mm and height 60 mm rests on one of its base side on the H.P. inclined at 30° to the V.P. Its axis is inclined at 45° to the H.P. Draw its projections.
3. A square pyramid of base side 40 mm and axis 55 mm is resting on one of its triangular faces on the H.P. A vertical plane containing the axis is inclined at 45° to the V.P. Draw its projections.
4. A hexagonal pyramid of base side 30 mm and axis 60 mm, has an edge of its base on the ground inclined at 45° to the V.P. and the axis is inclined at 30° to the H.P. Draw its projections.
5. A hexagonal pyramid of base side 30 mm and axis 50 mm, rests on one of its base corners on the ground with axis inclined at 45° to the H.P. Draw its projections when a vertical plane containing the axis and the corner that lies in the H.P. is inclined at 30° to the V.P.
6. A hexagonal pyramid of base side 30 mm and axis 60 mm has one of its slant edges on the H.P. and inclined at 45° to the V.P. Draw its projections when the base is visible.
7. A hexagonal pyramid of base side 30 mm and axis 60 mm rests on an edge of base on the H.P. with the triangular face containing that edge perpendicular to the H.P. and parallel to the V.P. Draw its projections so that the base is visible.
8. A pentagonal pyramid of base side 30 mm and axis 60 mm rests on a corner of its base on the H.P. such that its apex is 55 mm above the ground. A vertical plane containing the corner of the base that lies on the H.P. and the axis is inclined at 30° to the V.P. Draw its projections.
9. A pentagonal pyramid of base side 30 mm and axis 60 mm rests on an edge of its base on the ground so that the highest point of the base is 20 mm above the ground. Draw its projections when a vertical plane containing the axis is inclined at 30° to the V.P.
10. A pentagonal pyramid of base side 30 mm and axis 60 mm is held on a corner of its base on the ground and the triangular face opposite to it is horizontal. Draw the projections of the pyramid when the edge of the base contained by this triangular face is inclined at 60° to the V.P. and the apex is towards the observer.
11. A cylinder of base diameter 50 mm and axis 65 mm rests on a point of its base circle on the H.P. Draw its projections when the axis is inclined at 30° to the H.P. and top view of the axis is perpendicular to the V.P.
12. Draw the projections of a cube of edge 40 mm resting on one of its corners on the H.P. with a solid diagonal perpendicular to the V.P.
13. A square pyramid of base side 40 mm and axis 60 mm is freely suspended from a corner of its base. Draw its projections when the axis as a vertical plane is inclined at 45° to the V.P.
14. A cone of base diameter 50 mm and axis 60 mm is freely suspended from the midpoint of a generator. Draw its projections when the top view of that generator is inclined at 45° to the reference line and apex is nearer to the observer.
15. A square pyramid of base diagonal 50 mm and axis 60 mm is titled until the top view of the base appears as a rhombus having one of the diagonal twice of the other. Draw its projections when the axis as a vertical plane is inclined at 45° to the V.P.

16. A tetrahedron of edge 70 mm is resting on one of its edges in the H.P. which is inclined at 45° to the V.P. and a face contained by that edge is inclined at 30° to the H.P. Draw its projections.
17. A frustum of a pentagonal pyramid of base edge 40 mm, top edge 20 mm and axis 60 mm is resting on its face on the H.P. Draw its projections when an edge of its base is parallel to the V.P. and the small base is towards the observer.

An Element of the Solid in the V.P

18. A pentagonal prism of base side 30 mm and axis 60 mm has an edge of its base in the V.P. and inclined at 45° to the H.P. Its axis is inclined at 30° to the V.P. Draw its projections.
19. A hexagonal prism of base edge 30 mm and axis 70 mm has an edge of its base in the V.P. and inclined at 60° to the H.P. Draw its projections, when the edge of the other base farthest away from V.P. is at a distance of 85 mm from the V.P.
20. A square pyramid of base side 40 mm and axis 60 mm has a corner of its base in the V.P. A slant edge contained by that corner is inclined at 45° to the V.P. Draw its projections when a plane containing the slant edge and the axis is inclined at 45° to the H.P.
21. A cone of base diameter 50 mm and axis 60 mm has one of its generators in the V.P. and inclined at 30° to the H.P. Draw its projections when the apex is 15 mm above the H.P.

Condition of Apparent Angle

22. A hexagonal prism of base edge 30 mm and axis 60 mm rests on one of its base edges on the H.P. such that the axis is inclined at 30° to H.P. and 45° to the V.P. Draw its projections.
23. Draw the projections of a cone of base diameter 50 mm and axis 60 mm resting on a point of the base circle on the ground with axis inclined at 30° to the H.P. and (a) 45° to the V.P., and (b) the top view of the axis inclined at 45° to the V.P
24. A square pyramid of base side 40 mm and axis 50 mm has a triangular face on the ground and the axis inclined at 45° to the V.P. Draw its projections.
25. A pentagonal prism of base side 25 mm and axis 60 mm has its axis inclined at 60° to the H.P. and 30° to the V.P. The farthest shorter edge is parallel to and 90 mm above the H.P. while the nearest corner is 10 mm in front of the V.P. Draw its projections.
26. A cylinder of base diameter 50 mm and axis 70 mm has a point of its base circle in the V.P. Its axis is inclined at 30° to the V.P. and 45° to the H.P. Draw its projections.

An element of the solid in the H.P.

27. A pentagonal pyramid of base side 30 mm and axis 70 mm has one of the corners on the ground and its axis inclined at 45° to the H.P. A vertical plane containing the axis and that corner is inclined at 30° to the V.P. Draw its projections.
28. A hexagonal prism of base side 30 mm and axis 70 mm has an edge of the base parallel to the H.P. and inclined at 45° to the V.P. Draw its projections when its axis is inclined at 60° to the H.P.

29. Draw the projections of a cone of base diameter 50 mm and axis 70 mm, resting on a point of its base circle on the ground such that its axis is inclined at 30° to the H.P. and the top view of the axis is inclined at 45° to the V.P.
30. Draw the projections of a hexagonal pyramid of base side 30 mm and axis 65 mm, standing on an edge of the base in the H.P. inclined at 45° to the V.P. The slant face containing that edge is inclined at 60° to the H.P.
31. A triangular prism of base side 50 mm and axis 70 mm is resting on a corner of its base on the H.P. with a longer edge containing that corner inclined at 45° to the H.P. The vertical plane containing that edge and the axis is inclined at 30° to the V.P. Draw its projections.
32. A hexagonal pyramid of base edge 30 mm and axis 65 mm has a triangular face on the H.P. and the edge of the base containing that face makes an angle of 30° with the V.P. Draw its projections.
33. A hexagonal prism of base side 30 mm and axis 70 mm is resting on one of the edges of its base in the H.P. inclined at 60° to the V.P. The base is inclined at 60° to the H.P. Draw its projections.
34. A pentagonal pyramid of base side 30 mm and axis 60 mm has one of the edges of the base in the H.P. and parallel to the V.P. The solid is tilted in such a manner that the highest point of the base is 40 mm above H.P. Draw its projections.
35. A square pyramid of base side 50 mm and axis 60 mm has one of its triangular faces on the H.P. and a slant edge containing that face is parallel to the V.P. Draw its projections.
36. A square pyramid of base side 40 mm and height 70 mm has one of its triangular faces parallel to and 25 mm above H.P. and the shortest side of that face is inclined at 60° to the V.P. Draw its projections when the base is visible.
37. A cone of base diameter 60 mm and axis 80 mm rests on a point of its base circle on the ground with a generator normal to the H.P. Draw its projections when plan of the axis is inclined at 30° to the V.P.
38. A hexagonal pyramid of base side 30 mm and axis 75 mm has one of its triangular faces perpendicular to H.P. and inclined at 45° to the V.P. Draw its projections when the base side of the vertical face is on the H.P.
39. A cone of base diameter 50 mm and axis 70 mm is resting on one of its generators on the ground and inclined at 30° to the V.P. Draw its projections when the apex is nearer to V.P. than the base.
40. A hexagonal pyramid of base side 30 mm and axis 75 mm has one of its slant edges on the H.P. and vertical plane containing that edge and the axis is inclined at 30° to the V.P. Draw its projections when apex is 15 mm in front of the V.P.
41. A cube of side 50 mm has a corner in the H.P. Draw its projections when a solid diagonal is parallel to the H.P. and inclined at 30° to the V.P.
42. Draw the projections of a hexagonal prism of base side 25 mm and axis 65 mm, resting on the ground on one of its corners with a solid diagonal perpendicular to the V.P.
43. A tetrahedron of edge 70 mm has an edge in the H.P. and inclined at 45° to the V.P. A face containing that edge is perpendicular to the H.P. Draw its projections.
44. Draw the projections of a pentagonal prism of base side 25 mm and axis 65 mm, hanging from the roof of a building through its corner in such a way that a vertical plane containing that corner is inclined at 45° to the V.P.
45. A hexagonal pyramid of base side 30 mm and axis 70 mm is suspended freely from one of its base corners such that a vertical plane containing the axis is inclined at 45° to the V.P. Draw its projections.

46. A cone of base diameter 50 mm and axis 65 mm is freely suspended from a point of its rim such that the top view of the axis is perpendicular to V.P. and the apex is towards the observer. Draw its projections.
47. A square pyramid of base side 50 mm and axis 60 mm is freely suspended by a string tied at the mid - point of a side of its base. Draw its projections when a vertical plane containing the axis is inclined at 45° to the V.P.
48. A hexagonal pyramid of base edge 30 mm and slant edge 75 mm is resting on an edge of base on the ground in such a way that the edge of the base on which it rests is inclined at 45° to the V.P. and the base itself is inclined at 60° to the H.P. Draw its projections.
49. A square pyramid whose faces are isosceles triangle of base 50 mm and altitude 70 mm is resting on one of its faces on the H.P. such that the top view of the axis is inclined at 30° to the V.P. Draw its projections.
50. A pentagonal prism of base side 30 mm and height 60 mm rests on one of its base side on the H.P. such that the rectangular face contained by that edge appears as a square of 30 mm side in the top view. Draw its projections when the edge on the H.P. is inclined at 30° to the V.P.
51. A square pyramid of base edge 40 mm and axis 60 mm is resting on an edge of base on the H.P. such that the triangular face contained by that edge appears as an equilateral triangle in the top view. Draw its projections when the edge on the H.P. is inclined at 30° to the V.P.
52. A hexagonal pyramid of base side 30 mm and axis 60 mm rests on an edge of base on the H.P. such that the face contained by that edge appears as a straight line parallel to the reference line in the top view. Draw its projections. [Hints: Solution same as Fig. 11.39]
53. A tetrahedron ABCV of edge 60 mm has an edge AB on the ground and the face ABV is inclined to H.P. such that the plan of ABV is a right - angled triangle. Draw its projections when the edge AB is inclined at 30° to the V.P.
54. A bucket is in the form of the frustum of a cone has bottom diameter 75 mm, top diameter 30 mm and height 80 mm. The bucket is filled with water and then tilted through 45°. Draw its projections showing water surface in both views. The axis of the bucket is parallel to the V.P.
55. A crucible has top diameter 100 mm and bottom diameter 50 mm and height 80 mm. It is completely filled with molten metal and then tilted to pour the metal such that its axis is inclined at 45° to its initial vertical position. Show the surface of the metal in both the views. Wall thickness of the crucible is to be ignored.

An element of the solid in the V.P.

56. A hexagonal prism of base side 30 mm and axis 75 mm has an edge of the base on the V.P. and inclined at 30° to the H.P. The rectangular face containing that edge is inclined at 45° to the V.P. Draw its projections.
57. A pentagonal pyramid of base edge 30 mm and axis 70 mm has an edge of its base in the V.P. and parallel to the H.P. such that the triangular face contained by that edge is inclined at 45° to the V.P. Draw its projections.
58. A pentagonal pyramid of base side 30 mm and axis 70 mm has a triangular face in the V.P. and a slant edge containing that face is parallel to H.P. Draw its projections.
59. A square pyramid of base side 40 mm and axis 75 mm has a triangular face in the V.P. and an inclined plane containing the axis is inclined at 45° to the H.P. Draw its projections when its base is closer to the H.P. than the apex.

Condition of apparent angle

60. A square prism of base side 40 mm and axis 65 mm is resting on an edge of its base on the H.P. with axis inclined at 45° to the H.P. and 30° to the V.P. Draw its projections.
61. Draw the projections of a cone of base diameter 30 mm and axis 65 mm when it is resting on one of its generators on the ground and (a) that generator is inclined at 30° to the V.P., (b) the axis is inclined at 30° to the V.P.
62. Draw the projections of a pentagonal prism of base side 30 mm and axis 60 mm, resting on one of its edges of the base on the ground with the axis inclined at 30° to the H.P. and (a) 60° to the V.P., (b) top view of the axis is inclined at 60° to the V.P.
63. A cone of base side 60 mm and axis 75 mm is placed with its apex on the reference line and the axis is inclined at 30° to the H.P. and 60° to the V.P. Draw its projections.